ACMANTv6.0 homogenization software Manual

Servei Meteorològic de Catalunya

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Content

1. Introduction

General properties

ACMANT (Applied Caussinus – Mestre Algorithm for homogenizing Networks of climatic Time series) is known about its outstanding accuracy among tested homogenization methods (Guijarro, 2019; Domonkos et al., 2021; Guijarro et al., 2023). This method is suitable for the homogenization of several surface climate variables, namely temperature (TT), precipitation amount (RR), relative humidity (HH), wind speed or wind gust (FF), sunshine duration or radiation (of any kind) (SS) and atmospheric pressure (PP). All these variables can be homogenized either in daily or monthly scales. ACMANT can be run either in automatic or interactive mode, and metadata can be used together with the statistical procedure. Some further important properties of the method:

(i) ACMANTv6 (hereafter ACMANT) assesses the size of sudden shifts (breaks) in the section means, indicates and treats outlier values, provides the homogenized time series and fills data gaps (optional) with interpolated values. For TT, HH, FF, SS and PP, the homogenization of section means is based on the concept that inhomogeneities result in linear biases. Although the linearity is not perfect, the use of linear model gives good results in the homogenization of annual and multiannual means and linear trends. In RR homogenization, biases are supposed to be multiplicative.

(ii) Seasonally varying biases may be treated a) with the inclusion of a sinusoid model of seasonal changes (S-model), b) with individually estimated monthly correction terms (for irregular seasonality, I-model), or c) with the exclusion of the seasonal variation of correction terms (presumption of insignificant seasonal variation, flat, F-model). Users must choose between these three options (see guide in Sect. 4.2 (vii)). When the source area of the input dataset covers regions for which varied inhomogeneity seasonality models are favourable, the division of the initial dataset is recommended, since in ACMANT varied seasonality models cannot be applied within one homogenization procedure.

(iii) ACMANT homogenizes the probability distribution for TT, HH, FF, SS and PP when the input data set consists of daily data.

(iv) ACMANT includes an automatic networking procedure, therefore the spatial similarity in the climate of the source area of the input data is not required (with the exception mentioned in (ii)). Input datasets may have any size between 4 and 5000 time series. For input datasets of larger than 22 time series, or those including time series with lower spatial correlations than the minimum threshold, the software divides the input dataset to the set of networks of sufficiently correlating time series (multi-network homogenization). Each network includes one candidate series and a limited number of (max. 98, but usually less than 50) neighbour series. When the input dataset contains ≤ 22 sufficiently correlating time series, ACMANT performs 1-network homogenization.

(v) ACMANT removes automatically the physically impossible values from the input, and users may define thresholds for climatic outliers. Beyond the control of climatic outliers, ACMANT offers the filtering of spatial outlier values with monthly input, and the filtering of short-term spatial outlier periods with either daily or monthly input.

(vi) Interrupted multi-network homogenization can be continued without repeating the homogenization of the series whose esults have once been produced.

(vii) Input time series may cover varied time periods, but strict rules must be followed in their preparation.

(viii) Metadata can be used either in interactive or automatic modes. Regarding the automatic mode, the list of metadata dates must be prepared in the prescribed form. If metadata list is not provided and automatic homogenization is opted, the program will run without metadata use.

(ix) The software offers default solutions regarding several details of the homogenization, but users may select personalized options. The list of optionally alterable characteristics are as follows:

- Network structures (in multi-network homogenization);
- Minimum threshold (global) for spatial correlations (default = 0.4);
- Minimum threshold (network-specific, in multi-network homogenization) for spatial correlations (default = global threshold);
- Thresholds for climatic outliers (default = thresholds of physically possible range)
- Time series subjected to homogenization (default = all);
- Time series subjected to PDF homogenization (default = all);
- Time series subjected to interactive homogenization (default = none);
- Items of homogenization output;
- Gap filling in the homogenized time series

Any input dataset for ACMANT should be free of strikingly large, climatically impossible outlier values, values generated by spatial interpolation or by a previous homogenization procedure in which spatial comparisons were applied. Input datasets should preferably contain all the existing observed climatic data of the regions under study, as the accuracy of homogenization increases with the spatial density of data.

Changes in ACMANTv6 in comparison to ACMANTv5.3

The main novelty of version 6 is that it includes the homogenization of the probability distribution (PDF), which procedure (HPDTS = Homogenization of the Probability Distribution for Time Series) is described in Domonkos (2025a). One principle of HPDTS is that multiple inhomogeneities of the PDF are corrected by a modified version of Benova (Domonkos and Joelsson, 2024), and that modified version is called Cenova (Domonkos 2025a, 2025b). Before applying Cenova, low frequency climate variations are removed from the series, and they are added back once the corrections of Cenova has been executed. In the development of the present software, some further changes have been applied to the model described in Domonkos (2025a), the most important ones are:

(i) Annual means for each percentile ranges are homogenized with Benova in order to obtain more accurate estimates of the low frequency changes. The accuracy of low frequency change estimations is important, since that impacts the accuracy of the Cenova results, and thus it impacts the accuracy of the final homogenization results.

(ii) Climatic variables are sorted into three groups: Group A: inhomogeneity biases are presumed to be dependent on the absolute value of the variable; Group B: inhomogeneity biases are presumed to be anomaly and season dependent where “anomaly” means the deviation from the long-term seasonal mean; Group C: PDF-dependent inhomogeneity biases cannot be estimated with sufficient reliability. Although the type of climatic variable is not the only factor impacting the correctness of the described sorting, in ACMANT the sorting depends only on the climate variable: Group A: FF, HH and PP. Group B: TT and SS. Group C: RR. This means that no PDF homogenization is performed for preciptation. On the other hand, homogenization of PDF is performed only when the input data are of daily resolution, and the change of PDF is found to be significant for at least 1 detected break of the section means.

Further new features of ACMANTv6 is that the method allows more kinds of user interventions, and it can generate more kinds of optional output items. The related issues are partly re-structured also for the homogenization of monthly series and the homogenization of RR.

Users can find details of the theoretical content of ACMANT in the publications of Domonkos (2020, 2021,2022, 2025a). If you have questions, comments or any unexpected experience in using ACMANT, please do not hesitate to contact with the method developer:

Peter Domonkos
dpeterfree@gmail.com
www.acmant.eu

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2. Structure of directories and files

The software is put into a directory named “ACMANTv6.0”. In its main directory one can find the main executive file “ACMANT6run.BAT” together with 4 subdirectories (hereafter they are referred to as directories) and the file of this guide. The BAT file of the main directory opens an executive file of directory “Works”, which manages the homogenization procedure using 30 further BAT files and also 26 executive files and 2 text files.

Directory “Input” is for the deposition of input dataset and possible request files (explained later), while the homogenization products will appear in directory “Output”.

Directory “Partial results” is for keeping some partial results of the homogenization procedure there. Users use partial results when they opted for interactive homogenization.

Within directory Works, one can find several files and two subdirectories: “Auxiliary”, and “Subnetws”. In homogenizing datasets including more than 22 time series or pairs of time series with lower spatial correlation than the defined threshold, the software performs automatic networking whose results are kept in Subnetws (multi-network homogenization). Users may edit the content of Subnetws, but they should not do any change in other parts of directory Works. When a new run of ACMANT starts, the content of Subnetws is automatically emptied.

3. Preparation of input data

Before starting a homogenization procedure with ACMANT, users should prepare the data in the required format and put them into directory Input. It is strongly recommended to remove any other files from “Input” (e.g. the input data of an earlier homogenization) before this step. In Sects. 3.1 – 3.4, the rules for the preparation of input climatic datasets are presented. Note that ACMANT can be applied if the amount and temporal compactness of spatially fairly comparable data reaches certain (rather low) thresholds. The thresholds are shown in Sect. 3.5. Sect. 3.6 presents the rules for preparing metadata lists. Users may change default parameterization with showing the requested parameters in request files. The rules for editing such files are shown in Sects. 3.7 – 3.11.

All kinds of input information including metadata file (optional) and request files (optional) for personalized parameterization must be put into directory Input before starting the homogenization procedure.

3.1. Rules for all kinds of input climate data files

i) Number of time series per dataset is between 4 and 5000. As a general rule, a time series or its section will be homogenized if it has at least 3 neighbour series with temporally coincidental periods of observed data and sufficient spatial correlations (default threshold = 0.4) with the candidate series. A neighbour series or its section is usable only when that can be homogenized based on the same conditions as which are prescribed for the candidate series.

ii) Length of time series (may include data gaps): between 10 years and 200 years.

iii) Time series are separated into distinct files according to observing stations. In other words: 1 file is for 1 station time series only.

iv) Each file must contain a headline with the name of the station or other station identifier. There is no restriction for the length of the headline, but only its first 40 characters will be usable for identifying the station.

v) Default units of climatic elements: °C for temperature, mm for precipitation, % for relative humidity, hour for sunshine duration, m/s for wind speed and hPa for atmospheric pressure. You are allowed to use other units (except for precipitation) with the following precautions: a) It is indispensable to use the same unit throughout a given homogenization process. b) The wrong use of units might cause overshooting of physical thresholds inbuilt in programs or overshooting of user defined climatic thresholds even when the input data is correct. c) Climatic values below -998 are considered missing data, and those which are higher than 9999 cannot be treated by the software.

vi) Each line (after the headline) contains a pre-determined number of values (i.e. date identifier(s) and 12 monthly values in monthly datasets or 28-31 daily values in daily datasets). The values in the same line are separated with one or more space characters. TAB also can be applied for separation (optional). The use of other characters (comma, semicolon, etc.) is not allowed. Date identifiers are shown with whole numbers, while the data of climatic variables are usually presented with 1 decimal preciseness, but note that other forms are also accepted. Decimals are separated by dot from the other digits. E.g. value – 4°C can be presented in any form of – 4, – 4.0 or – 4.00.

vii) Missing values are represented by –999.9.

viii) Trace precipitation must be coded with 0.

ix) Gap filling with missing value code

Input time series must be temporally continuous from the first calendar day (or month, in case of monthly input) of the year of the first observed data until the last calendar day (month) of the year of the last observed data, although some exceptions are allowed (Sect. 3.4).

Best Site Station

```
1951  -999.9 -999.9    4.5    7.3     15.6    17.6    -999.9 -999.9 -999.9 -999.9 -999.9 -999.9
1952     -1.5    0.0    7.1    9.8    10.9    15.6     19.8    20.3     13.8  -999.9    8.1     0.4
1953 -999.9 -999.9 -999.9 -999.9 -999.9 -999.9 -999.9 -999.9 -999.9 -999.9 -999.9 -999.9
1954 -999.9 -999.9 -999.9   8.6    14.2    16.8      18.8    18.5     16.4      9.5     3.6    -0.2
……………….
……………….
……………….
```

In Best Site Station the temperature observation started in March 1951, but there are several gaps in the first few years. Please remember that it is not allowed to jump over the year 1953, in spite of that there was no observation at all in that year. It is also important to note that the time series starts with the January of the first observed data (missing data codes for January and February of 1951).

3.2. Specifics for monthly input datasets

i) Time-resolution is monthly.

ii) Name of files: “ANAMEJJJJ.txt”.

- “ANAME” is the name of the dataset. This substring is always 5-character wide and may contain any alphanumeric characters.
- “JJJJ” is the serial number of time series, it always consists of 4 digits.
- “.txt”: these 4 characters are fixed, they are always “.txt”

Example: If the name of the dataset is “Lucky” and it consists of 275 time series, then the names of the time series must be:

Lucky0001.txt
Lucky0002.txt
Lucky0003.txt
……………..
……………..
Lucky0274.txt
Lucky0275.txt

iii) Each line (after the headline) is for one calendar year. Thus the number of lines in file is the same as the number of years in the time series, plus the headline.

iv) Each line consists of 13 values: the first is the calendar year, while the other twelve are the monthly climatic values for that year in their natural order (from January to December).

3.3. Specifics for daily input datasets

i) Time-resolution is daily.

ii) Name of files: “ANAMEJJJJd.txt”.

- “ANAME” is the name of the dataset. This substring is always 5-character wide and may contain any alphanumeric characters.
- “JJJJ” is the serial number of time series.
- “d.txt”: these 5 characters are fixed, they are always “d.txt”

Example: If the name of the dataset is “Smart” and it consists of 7 time series, then the names of the time series must be:

Smart0001d.txt
Smart0002d.txt
………..
Smart0007d.txt

iii) Each line (after the headline) is for one calendar month. The first line (after the headline) is for the January of the first year, while the last line is for the December of the last year. Thus the number of lines in file is the same as the number of years in the time series multiplied by 12, plus the headline.

iv) Each line consists of 2 date identifiers and 28-31 values. The date identifiers are the calendar year and the serial number of calendar month. The other values are the observed climatic data in their temporal order (from the first day to the last day of the month). If a line contains less values than the number of days in the month, the program will stop with an error message. By contrast, if lines contain more values than the number of days in the month, the program will not stop, but the data at the end of lines will not be read by the program.

v) Example of daily precipitation series with observed data between 5 May 1954 and 29 November 2006. Observation was suspended between 30-Dec-1954 and 03-Jan-1955.

```
Nice Park City
1954  1  -999.9 -999.9 -999.9 -999.9 -999.9 -999.9……..-999.9 -999.9 -999.9 -999.9
1954  2  -999.9 -999.9 -999.9 -999.9 -999.9 -999.9……..-999.9
1954  3  -999.9 -999.9 -999.9 -999.9 -999.9 -999.9……..-999.9 -999.9 -999.9 -999.9
1954  4  -999.9 -999.9 -999.9 -999.9 -999.9 -999.9……..-999.9 -999.9 -999.9
1954  5  -999.9 -999.9 -999.9 -999.9    1.4    0.0 ……   16.2    0.9    0.0    0.1
1954  6     0.0    0.0    7.3   18.2    0.0    0.4 …....  0.0    0.0   10.3
1954  7     2.2    0.0    0.0    0.0    0.0    0.0 ……    0.0    0.0    2.9    0.0
1954  8     0.0    0.0    1.1    0.0    0.0    6.4 ……    0.0    0.0    0.0    0.0
1954  9     0.0    3.3   21.6   12.2    0.5    0.0 ……    0.0    0.1    0.0
1954 10     0.0    0.0    0.0    0.7    0.6    0.0 ……    5.2    3.4    0.7    3.2
1954 11     0.2    0.0    2.9    0.0    0.0   10.2 …….   0.0    0.0    0.0
1954 12     7.4    0.1    0.1    0.0    0.3    0.0 …….   1.0    0.0 -999.9 -999.9
1955  1  -999.9 -999.9 -999.9    0.0    0.0    0.7 ……    0.0    9.1    1.4    0.0
1955  2     3.0    0.0    0.0    0.0    0.2    0.0 ……    0.0
1955  3     0.0    2.7    2.6    0.1    0.4    0.0 ……    0.0    0.0   13.1    0.0
…………………………………………………………………………………………..
…………………………………………………………………………………………..
…………………………………………………………………………………………..
2006  9     0.0    0.0    1.6    0.0    2.2    6.0 …...   0.7    0.0    0.0
2006 10     1.3    0.0    1.2    0.1    4.4    0.0 …...   0.0    0.0    0.0    0.0
2006 11     0.0    0.0    5.1    0.0    0.0    0.4 …...   0.0    0.0 -999.9
2006 12 -999.9 -999.9 -999.9 -999.9 -999.9 -999.9…….-999.9 -999.9 -999.9 -999.9
```

3.4. Exceptions when input time series are allowed to be fragmented

Before starting the homogenization, the user must define the target period of the homogenization (Sect. 4.2). The input data out of this target period are allowed to be fragmented if they comply with the following conditions.

i) Before the beginning of the target period: If data for at least one date is included from a given year, then all the dates of that year (i.e. all the months or all the days according to the resolution of the dataset) must be represented either with observed data or with missing data code. However, years without any observed data can be jumped out.

ii) After the target period: If the input time series includes the last year of the target period, then the reading of that file will be terminated with the data of that year, and thus anything might be present in the later part of the file. However, if the series of continuous data terminates before the last year of the target period, the file must be ended together with the end of the series of continuous data.

Note that if you use any of these alleviations, this might cause problems if you repeat the homogenization of the same time series with varied target periods.

3.5. Conditions for treatment and homogenization by ACMANT

i) Requirements for the treated period of a time series

Sometimes time series include large missing data fields at the beginning and/or at the end of their period. During the homogenization procedure, the input time series are split into three sections: a treated section (referred to as treated period), and two not treated sections, one in the beginning part and another one in the ending part of the time series. Not treated sections remain out of most steps of the homogenization procedure, yet inhomogeneity adjustments and data completion by spatial interpolation may be applied in them. When the number of missing values is low, the whole period of the time series is treated period. The ratio of missing values within the treated period is limited according to the following rules:

a) In the first *k* year and last *k* year of the treated period the expected ratio of available observed data is higher than 25% for any *k* between 1 and 20, or between 1 and the length of the treated period if the latter is the shorter. For instance, if the first year is 1953 and the last year is 2002, the minimum number of observed monthly data for periods 1953-1953, 1953-1954, 1953-1967 and 2000-2002 are 4, 7, 46 and 10, respectively. Also, at least two neighbour series with sufficient ratio of observed data are needed.

b) The minimum length of a treated period is 10 years.

c) The minimum number of observed monthly data is 90 (so that even a time series of only 10-year length may include data gaps).

d) For each 3-month season like January-February-March, February-March-April, etc.: the minimum number of observed monthly values is 7.

e) Even when the input data is daily, the adequacy of the amount and temporal coherence of data is checked on monthly scale, according to the rules listed above. For this check, the status of monthly climatic value is “observed” when there are no more than 7 missing daily values in month and the status is “missing” in the reverse case. The status of monthly precipitation total is “observed” only if all the daily data are observed.

f) ACMANT excludes years with data in less than 3 stations of a network. More precisely, for each year of a treated period, at least 9 months (not necessarily the same set of months) with status “observed” are needed for at least 3 stations of a network. The exclusion of years might make short or fragmented time series to be fully untreated.

In case of too few or too fragmented data, time series might remain fully untreated. Note, however, that the homogenization procedure does not stop for the exclusion of one or more time series.

ii) Requirements for homogenizing series or their sections

a) If a station series (or its section) does not have at least 3 neighbour series with sufficient spatial correlation, that series (or its section) will not be homogenized, but otherwise the program will run normally. If none of the series can be homogenized, the program will stop with the adequate control message. Note that when no section of a candidate series can be homogenized, neither inhomogeneity adjustments nor data completion are applied to that series.

b) When there is a section of candidate series without 3 neighbour series, that section will not be homogenized. Rarely, both before and after an excluded section the number of comparable time series is higher, thus two (or more) distinct sections of a candidate series could be homogenized. However, in such cases only one section, i.e. the late section will be homogenized, even if a longer section could have been homogenized in the early part of the candidate series. If more than one separate homogenizable sections are found by ACMANT, then this information will appear in the output file “Control message.txt”.

The section of time series which can be homogenized according to the presented conditions is referred to as the homogenized period of the time series. Often the whole period of the input time series is homogenized period. Not treated sections cannot be part of the homogenized period. The use of data falling within the treated period but out of the homogenized period is strongly limited in the homogenization procedure.

3.6. Metadata list

Users may provide a list of metadata dates before starting a homogenization procedure. When such a list is provided, the homogenization procedure will use the metadata dates in combination with the break dates detected by statistical methods.

All the metadata of a given input dataset must be shown in one common file. Its format is the same for daily and monthly climatic datasets, since metadata are always shown with daily preciseness. Even when you do not know the exact date, the estimated date must be shown with daily preciseness.

The name of metadata list file: “ANAMEmeta.txt”, where

- “ANAME” is the dataset name (the same as for the climatic data)
- “meta.txt” is constant

The size of this file depends on the number of known metadata dates, and there is no need to order the pieces of metadata either according to time or station serial numbers.

In each line of the file, 4 whole numbers must be shown, they are from left to right: a) serial number of station, b) day, c) month, d) year. For instance, LUCKYmeta.txt has 20 time series, but only 4 metadata:

```
14  31  11  1970
14  12  4  1970
5  31  12  1988
9   7   11  1963
```

The example shows that more than one metadata may occur for the same year and same station, but note that too high metadata frequency is generally neither recommended nor reasoned. The maximum number of pieces of metadata for a station is 40, while for an entire input dataset 40,000. Warning: Any erroneous date will stop the program with the relevant error message. However, when metadata file is not provided, it does not cause problem, and the program will run without metadata use.

3.7. Thresholds for climatic outliers

ACMANT can be run with the inbuilt physical thresholds, or user defined climatic thresholds can be applied.

i) Inbuilt thresholds
temperature: -98 and 60
daily precipitation: 0 and 1500
monthly precipitation: 0 and 6000
relative humidity and wind speed: 0 and 100
sunshine and radiation, daily: 0 and 24
sunshine and radiation, monthly: 0 and 744
atmospheric pressure: 0 and 1099

ii) User defined thresholds
User defined thresholds overwrite the inbuilt threshold values. While the inbuilt thresholds are season-independent, user defined thresholds consist of monthly low threshold and high threshold values for each calendar month. These thresholds are written

to the request file “Thresholds.txt”. The file must include 12 lines, with the serial number of calendar month and the monthly low threshold and high threshold values in each line. An example of Thresholds.txt for the homogenization of temperature maximums in a Mediterranean region:

```
1  -10   35
2  -10   35
3  -10   40
4   0     50
5   5     50
6   10    55
7   10    55
8   10    55
9   10    50
10   5    45
11   -5    40
12   -10  35
```

Note that lower (higher) threshold values than -997.9 (9999) cannot be defined, and in precipitation homogenization negative values are not allowed.

When the use of inbuilt physical thresholds of the software is preferred, the input package should not contain Thresholds.txt.

3.8. Spatial correlation

Two kinds of spatial correlations are used in ACMANT, one is for homogenization, and another is for gap filling. The ways of their calculation differ (Domonkos, 2020). For gap filling, the minimum threshold is 0.4 and users cannot modify it. For homogenization, the spatial correlations are calculated from de deseasonalised values of monthly increment series (see more in Domonkos, 2020). In precipitation homogenization the spatial correlations are calculated from the re-scaled data by a semi-logarithmic transformation (Domonkos, 2020). The default minimum threshold for homogenization is 0.4, but users can alter this either for the entire dataset, or for some selected networks. If you would like to use another minimum threshold for the entire dataset, then you must show it in the relevant request file.

The filename is constant: “Rth.txt”. The requested spatial correlation threshold must be written into its first line without adding any other thing. The program accept any value between 0.1 and 0.99. When Rth.txt is not shown in directory Input, or a valid correlation threshold cannot be found in that, the program will use the default correlation threshold.

In interactive mode, users may define individual correlation thresholds for selected networks, this will be shown in Sect 4.6.

3.9. Target series for homogenization

For relatively small datasets, ACMANT usually homogenizes all the series together without grouping them into networks.

When networks are constructed by the program, still all the series of a network are homogenized together, but in such networks the results are saved only for the candidate series of the network. It is because a principle of the network construction in ACMANT is that a network is optimal for one selected candidate series, and therefore the number of networks equals to the number of the time series of the input dataset (see also Domonkos, 2020).

In homogenizing datasets of larger than 22 time series (multi-network datasets), users may save time with defining which time series are expected to be homogenized when not all the series are needed to be homogenized. The serial numbers of the requested time series must be written into the request file “ANAMEtarget-homg.txt”. In this filename

- “ANAME” is the dataset name (the same as for the climatic data)
- “target-homg.txt” is constant

Each line of the file must contain one serial number. You may write the relevant serial numbers in any order. If the same serial number is shown more than once, this does not cause error. The maximum length of the file is 5000 lines. However, empty lines or invalid serial number of time series are not allowed.

Example: dataset WHITE includes 1622 time series, but you need the homogenization results only for series 7, 78, 922, 926, 1473 and 1496. Then WHITEtarget-homg.txt file must be created with the following content:

7
78
922
926
1473
1496

If WHITEtarget-homg.txt is not shown in directory Input, ACMANT will homogenize all the 1622 series. If you write 1622 into the first line of WHITEtarget-homg.txt and nothing else, then ACMANT will homogenize only the last series of the dataset.

Note that the request of the homogenization of any time series can also be defined either by “ANAMEtarget-pdfh.txt” file (Sect. 3.10) or “ANAMEedit.txt” file (Sect. 3.11). And note again when ANAMEtarget-homg.txt is not used, all the time series will be subjected to homogenization by ACMANT.

When ACMANT performs 1-network homogenization, “ANAMEtarget-homg.txt” will not be considered and all the time series will be homogenized.

3.10. Target series for the homogenization of probability distribution

If you would like to run ACMANT without applying probability distribution homogenization, you should create the request file “ANAMEtarget-pdfh.txt” where

- “ANAME” is the dataset name
- “target-pdfh.txt” is constant

The rules of the edition of this file is similar to those of “ANAMEtarget-homg.txt”: Each line contain the serial number of one time series of the dataset, the file may contain any number of lines up to 5000, the serial numbers can be written in any order, and possible repetitions of serial numbers (although unnecessary) do not cause homogenization errors. A specific of “ANAMEtarget-pdfh.txt” is that 0 may be written in its first line. The meaning of that is that probability distribution homogenization is not required for any of the time series. If 0 is written into the first line, possible further lines of the file are left without consideration.

In case of 1-network homogenization (of daily series of relevant climatic variables), the probability distribution homogenization will be performed for all time series, except when 0 is written in the first line of the request file.

3.11. Target series for interactive homogenization

ACMANT allows user interventions to the network structures (in multi-network homogenization), and in some other steps of the homogenization procedure. However, to perform interactive homogenization, user should create request file “ANAMEedit.txt” and put it into directory Input. In this file

- “ANAME” is the dataset name
- “edit.txt” is constant

The rules of the file edition are similar to those of “ANAMEtarget-homg.txt” and “ANAMEtarget-pdfh.txt”. Each line of the file contains the serial number of one time series of the dataset, the serial numbers can be written in any order, and the file may contain any number of lines up to 5000.

If you want to homogenize each time series of a dataset in interactive mode, you should write all the serial numbers into the request file.

User may write 0 in the first line of “ANAMEedit.txt”. In such cases, ACMANT will only allow the user to modify the automatically constructed networks.

4. Running ACMANTv6

4.1. General conditions

The software package with its directory structure must be placed in your computer. Saving a copy is advised. The preferred environment is Windows, since the compiled FORTRAN programs of ACMANT are directly executable under Windows. Under Linux, de program can be run by “Wine” application, starting the run with the command “wineconsole ACMANT6run.BAT”.

The computational time demand widely varies: for small datasets allowing 1-network homogenization the run usually lasts less than 1 minute. In multi-network homogenization the typical time demand is from 1 hour to 1 day, but the homogenization of very large datasets (>1000 series) may last for weeks. In multi-network homogenization the time demand increases linearly with the number of time series and exponentially with the length of the target period. The homogenization of daily data needs 2-4 times longer time than the homogenization of monthly data. The network size has strong impact on the time consumption, this fact is recommended to be taken into account in case of manual network editions.

Even if you use automatic mode, you must introduce some parameters manually at the beginning of the homogenization procedure, these are described in Sects. 4.2 - 4.3. Thereafter the program can do everything without user assistance, and thus the content of Sects. 4.4 – 4.10 might be less interesting for users of the automatic mode.

4.2. Initiation I. Starting a new homogenization

Once the input dataset and supplementary input files have been prepared, they are placed to directory “Input”. Please check that directories “Input” and “Output” do not include data of earlier homogenizations.

The homogenization starts with clicking on “ACMANT6run.BAT”. The program will ask for typing some parameters on the screen. When the answer includes (or may include) letters, please introduce the answer from the left end of the line, without space characters.

i) Climatic element: Two characters. It is “TT” for temperature, “RR” for precipitation amount, “HH” for relative humidity, “FF” for wind speed or wind gust, “SS” for sunshine duration or radiation, and “PP” for atmospheric pressure.

ii) Temporal resolution: One character, “m” for monthly homogenization and “d” for daily homogenization.

iii) Name of the dataset: 5 characters, the same as which is included in the filenames of the input time series.

iv) Number of stations: The number of time series in the input dataset must be given here.

v) – vi) First calendar year of time series and number of years in time series: These two parameters determine the target period of homogenization. If all the input time series cover the same time period, it is straightforward to define that period as target period, but the period of observations might vary widely in some datasets. You are allowed to choose either shorter or longer target period than which is covered with input data, there are two requirements only: a) the length of target period must fall between 10 and 200 years; b) All dates of the metadata list (if metadata list is provided) must fall within the target period.

vii) Seasonality of inhomogeneities: It can be sinusoid ("S"), irregular ("I") or flat (F). This question appears in the homogenization of TT, HH, FF, SS or PP datasets, but does not appear in precipitation homogenization.

In the middle and high latitudes not only the mean climate, but also the inhomogeneities often have quasi-harmonic shape with modes around the solstices. In these regions, the selection of S-model is recommended for the homogenization of mean temperature, maximum temperature, relative humidity, sunshine duration, radiation and sea level pressure, while the I-model is recommended for the homogenization of minimum temperature and wind speed.

For tropical belt regions and regions of subtropical monsoon climate, the application of I-model is recommended for all climate variables except for precipitation.

The seasonality of RR inhomogeneities is discussed in (viii-ix).

viii) – ix) First and last snowy months: These questions appear only in the homogenization of precipitation (RR).

In large regions of the earth snow rarely or never falls. There the presumed seasonality is flat, and "0" must be answered to the question for snowy months. Where snow dominates over rains in a part of the year, the year should be divided to a rainy season and a snowy season with answering the first and last snowy months.

The average climatic characteristics of the area of the input dataset must be considered here, and months are coded with their serial numbers within a calendar year. For instance, if the first snowy month is December, and the last snowy month is March, the correct answer is "12" to (viii) and "3" to (ix). When the user defined season is shorter than 3 months, shorter than 3-month, the season length is extended by ACMANT with 2 months, by adding the adjacent calendar months before and after the user defined period.

x) Would you like outlier filtering?: This question appears only in monthly homogenization, and the answer is 1 character, "Y" if yes and "N" if no. The recommended response here is yes.

xi) Would you prefer default output package?: The answer can be "Y" or "N", and if "Y" is chosen, no further question is put. The default output package includes:

- homogenized time series in the same time resolution as that of the input time series;
- list of neighbour series and their spatial correlations with the candidate series;

- list of detected breaks and outliers
- only for time series of interactive homogenization: result tables provided for interactive Homogenization (Sect. 5.5).

A more detailed description of the default output package is presented in Sect. 5.1 and 5.2.

When the default output package has not been accepted, ACMANT puts further questions:

xii) Time resolution of homogenized time series. This question appears only when the input dataset is of daily resolution. “d”= daily, “m”= monthly, “2”= both kinds of output items are required.

xiii) Gap filling. “0”= never, “1”= within the homogenized period, “2”= completion of series for the target period, “3”= all kinds of output items of the previous three options.

xiv) Table of confidence indicators: “Y” if yes, “N” if no. Indicator values show for individual monthly or daily values if they are homogenized observed data, interpolated data, etc. The indicators are whole numbers between 0 and 9. Lower values (except for 0) mean higher reliability, see details in Sect. 5.4.

xv) Lists and tables of calculation results: “Y” if yes, “N” if no. This package includes various lists and tables (Sect. 5.5), which are generated without request when interactive homogenization is performed. If “Y” is chosen, the output will include such tables for all time series, while in case of “N”, they will be generated only for the time series subjected to interactive homogenization. When “Y” has been chosen, no further question will be put.

xvi) List of detected breaks and outliers. “Y” if yes, “N” if no. This item is a part of the default output package (xi) and the package of (xv), but you may request it here, if you have not accepted either (xi) or (xv).

xvii) List of neighbour series and their spatial correlations with the candidate series. “Y” if yes, “N” if no. This item is a part of the default output package (xi) and the package of (xv), but you may request it here, if you have not accepted either (xi) or (xv).

4.3. Initiation II. Continuation of a suspended homogenization

User may suspend the homogenization of multi-network datasets. When you want to continue the homogenization, click again on ACMANT6run.BAT, then the program will ask for the climatic element. To that question, “CC” should be responded, instead of the code of the climatic element. From this response ACMANT understands that the user wants to continue a suspended homogenization, and will ask you the serial number of the series in the homogenization accomplished last time. You can check that from directory Output and introduce the correct response. Then ACMANT will run without putting further questions. Note: The way of homogenization cannot be influenced by changing the content of directory Input during suspensions.

4.4. Networks

For large datasets, ACMANT automatically creates networks for each candidate series of the dataset. The selection of neighbour series is based on the spatial correlations and the coverage of the period of the candidate series (when some neighbour series are shorter than the candidate series or include data gaps). These automatically generated networks can be modified by the user. However, this is only the brief essence of networking, since ACMANT may use 3 differing network types in its procedure. The reason behind varying networks in the homogenization of a given candidate series is that the required number of neighbour series generally decreases with the gradual improvement of the data accuracy. Even in 1-network homogenization, 2 network types can be used, these things are explained below:

- Base network: This used in most steps of multi-network homogenization with ACMANT. A base network contains up to 22 sufficiently correlating time series, and may include further time series if the coverage of some sections of the candidate series needs higher number of neighbour series. Only neighbour series with sufficient spatial correlations (default threshold = 0.4) may be included. This network type is always generated in multi-network homogenization, i.e., when the input dataset contains more than 22 time series or contains at least 1 pair of time series whose correlation is lower than the correlation threshold. If the spatial correlation is incalculable for the lack of the sufficient number of data pairs, this has the same effect on networking as if the correlations would be too low. However, if a time series is left out of consideration from all the homogenization procedure for its insufficient data amount, this does not impact the way of the homogenization of the rest of the series.

- Extended network: This used in some early steps of the multi-network homogenization with ACMANT. When a large number of sufficiently correlating time series are available, the automatic network creation selects up to 29 neighbour series, and when the coverage of some sections of the candidate series is insufficient, the number of neighbour series can be even larger. Nevertheless, even when the period of observation is highly varied for individual time series, networks can rarely be larger than 50 time series. Note when the number of sufficiently correlating neighbour series is less than 22 for a given candidate series, the extended network is identical with the base network.

- Network for the homogenization of probability distribution, referred as also to small networks. This network consists of the candidate series and up to 10 neighbour series. ACMANT usually selects the best correlating neighbour series for each time series, exception can be that neighbour series with shorter than 20 years homogenized period may be excluded. This kind of network is individually edited by ACMANT for each time series even when “1-network homogenization” is performed.

The base networks and extended networks are generated before starting the homogenization of the time series, and they are put into directory Subnetws. In contrast, the small networks are generated during the homogenization within a base network. Subnetws is divided to sub-sub directories, as one specific base network is constructed for each candidate series of the dataset, and each of them has an own directory (network directory, hereafter). Network directory names show the serial number of the candidate series of the network. Inside a network directory, the candidate series and its neighbour series are present in the same form as in the input dataset, except for the filenames. The filenames of time series hold serial numbers specific for the network structure and independent from the serial numbers shown in the input dataset. Serial number 1 always belongs to the candidate series of the network. The other serial numbers show the order of selection of neighbour series during the network construction. Lower serial number indicates higher importance of the neighbour series for higher spatial correlation with the candidate series, or, for better coverage of the candidate series.

Each network directory includes a file “networksize.txt” presenting two numbers. The first number shows the number of time series of the base network, while the second number shows the number of series of the extended network, and the latter is identical with the number of files of time series which are put into the network. For instance, in a network where 30 time series can be found, “networksize.txt” may show: 24 and 30. This means that the first 24 time series compose the base network, while all the 30 time series compose the extended network.

4.5. Partial results and their edition

Partial results are tables showing some details of the homogenization of the actually examined network. Their purpose is to support interactive homogenization, although their content is output, even without interactive homogenization, when user answers positively the question “Lists and tables of calculation results?” at the beginning of homogenization (Sect. 4.2). Most (although not all) pieces of the partial results give information about all the time series of the actually homogenized network. When 1-network homogenization is performed, the time span of such partial results is from the earliest starting year of homogenized periods until the latest ending year of homogenized periods occurring in the network, while in multi-network homogenization it is identical with the homogenized period of the candidate series.

The partial results are put to directory “Partial results”. Some of them are editable by the users, while others serves only as supporting information for user’s decisions. ACMANT clearly indicates which file of the Partial results can be edited at a given phase of the homogenization procedure. If user don’t want to make editions he/she must simply click “C” (= continue). Generally, the format of editable files must be kept during editions in the format provided by ACMANT. For instance, if two pieces of information are separated by 1 blank line, then no more and no less than 1 blank line is expected there in the edited file. The number and position of space characters within a line are less strict, but the best is to maintain the format of the automatically generated file in all aspects. Logical connections between pieces of the information (when such occur) should also be treated carefully. After the editions has been finished at a given phase of the homogenization, “C” must be clicked. In the rest of Section 4 the user’s edition options and the related partial results will be detailed.

ACMANT skips the generation of some kinds of partial results when no usable results can be shown (e.g., when a time series cannot be homogenized).

4.6. User editions in the base networks and extended networks (optional)

This option is offered if “ANAMEedit.txt” is created with any acceptable content (Sect. 3.11). Two main kinds of editions for networks are possible:

- Users may add or remove time series into/from any network, except that the candidate series must remain the same in each network;
- Users may introduce network-specific correlation thresholds with putting “subrth.txt” files.

i) Adding or removing time series

Users might add or remove time series to / from a network for various reasons. For instance, additional time series might be added for increasing the weight of an underrepresented geographical direction, time series might be removed for their general quality problems, etc. In any case, users must assure that: (a) the list of the serial numbers of the time series must remain continuous; (b) the two numbers in file “networksize.txt” must show the number of series in the base network and extended network, respectively, so that that numbers should be adjusted when it is necessary for the increase or decrease of the number of time series.

Example: a base network contains 24 time series, while its extended version contains 30 series (ANAME01.txt, ANAME02.txt,…ANAME30.txt). If you remove the series with serial numbers 12 and 15, some files must be renamed. The simplest solution is if ANAME25.txt is renamed to ANAME12.txt, ANAME26.txt to ANAME15.txt, ANAME29.txt to ANAME25.txt and ANAME30.txt to ANAME26.txt. In this way the serial numbers will have the range from 1 to 28, the extent of the base network remains unchanged (24 series), while that of the extended network will be 28. Other solutions in renaming the files are also possible. Please, do not forget to adjust the content of “networksize.txt” accordingly.

Adding time series to a network is somewhat more complicated than removing time series, its reasons are presented in (ii).

ii) Network-specific correlation thresholds:

User can introduce network-specific correlation threshold by creating the file “Subrth.txt”, and putting it into the directory of the network. Similarly to “Rth.txt”, Subrth.txt should contain one only number indicating the correlation threshold in the first line of the file. For instance, lowering the correlation threshold allows to include time series which have somewhat lower correlation with the candidate series than those neighbour series which are selected by the automatic networking. Note, however, that changing the correlation threshold might impact in various ways the homogenization results of a candidate series, since in and ACMANT procedure all time series are homogenized together, and thus the change of the correlation threshold might influence the neighbour series selection for any time series of the network impacting indirectly the homogenization results of the candidate series.

On the other hand, adding a new neighbour series without changing the correlation threshold is ineffective if the selected neighbour series has lower spatial correlation with the candidate series than the global correlation threshold.

A further difficulty in adding new neighbour series to a network is that the format of time series files at this stage of the ACMANT procedure somewhat differs from the input data format. The format of newly added files should be the same as that of the automatically generated files, and users should treat the following issues:

- The time period of the series is fixed and identical with the target period defined at the start of running ACMANT.
- When the time series has metadata, the number of metadata dates must be shown in the line following the last line of the climatic data, then each metadata date must be written in separate lines, showing the year, month and day of date in this order and in the same line. Here, years are shown in different form from calendar years, i.e. the serial number of the year within the target period of the homogenization must be shown. When a time series does not have metadata, a 0 must be written into the first line following the table of the climatic data. An example: Let suppose that the precipitation observations of Nice Park City (shown in Sect. 3.3) have two metadata with dates 01-July-1987 and 15-May-1994, and the target period of homogenization is from 1954 to 2006. Then the data table edited for a network will end in this way:

```
2006 10    1.3     0.0     1.2     0.1     4.4     0.0  …...     0.0     0.0     0.0     0.0
2006 11    0.0     0.0     5.1     0.0     0.0     0.4  …...     0.0     0.0 -999.9
2006 12 -999.9 -999.9 -999.9 -999.9 -999.9 -999.9…….-999.9 -999.9 -999.9 -999.9
2
34  7  1
41  5  15
```

Note that the permitted maximum network size is 99 series, but the computation time for a network doubles with adding 5-7 time series to it, therefore automatically created networks should be enlarged only when the inclusion of additional neighbour series is clearly reasoned.

Let we turn back to the example network of ANAME (point (i)), whose base network has 24 time series and extended network has 30 time series by the automatic network generation of ACMANT. If you add a new neighbour series with serial number 31, that will be considered only in the extended network. For including the new series in the base network you should add to it a ≤ 25 serial number and rename some files to gain the continuous serial numbers from 1 to 31. File networksize.txt should be adjusted accordingly to the introduced changes.

4.7. User editions of the set of detected breaks (optional)

Users may edit the list of detected breaks only for the results of the first homogenization cycle. Such editions tend to be less effective than the network editions, since in the second and third homogenization cycles of ACMANT the automatic procedure may overwrite the user's contribution.

The editable file is PBRtypeANAMEJJJJ.res, where
"PBR" – constant
"type" – "Year" for annual mean, "S-Wd" for summer - winter difference when the sinusoid (S) model of the annual cycle of inhomogeneity biases is applied, "Rain" ("Snow") for precipitation total of rainy (snowy) season in precipitation homogenization.
"ANAME" – dataset name
"JJJJ" – serial number of the candidate series
".res" – constant

For a network of *N* time series, this file contains *N* blocks. Blocks are separated by one blank line. For each block:

- line 1: serial number of time series within the given network, homogenized period, station identifier
- line 2: number of detected breaks (*k*)
- lines after 2 until 2+*k*: serial number of break, year of the break (the end of the shown year must be considered), break size.

Users may alter the number of detected breaks or the year of the break position(s) for any time series, while possible editions for break sizes are ineffective. The format of the automatically generated file should be kept, but break size is not an obligatory (and in fact unconsidered) field. When user changes the detected number of breaks for a time series, the content of line 2 (number of detected breaks) of the block of the file must be corrected. Leaving 1 blank line between two consecutive blocks is expected.

Three kinds of non-editable partial results are presented together with the editable break list, their purpose is to support user's decisions. The additional files contain:

i) List of detected outliers in the first homogenization cycle;
ii) Table of pairwise comparison scores;
iii) Anomalies of annual values relative to the long-term mean values, calculated from outlier filtered but non-homogenized values.

In bivariate homogenization the editable file and type (iii) supporting files have two subtypes, i.e., result tables are shown for both variables. Bivariate homogenization is performed for a) S-model inhomogeneities, b) precipitation homogenization with alternating rainy and snowy seasons.

Note that in precipitation homogenization the logarithmic transformed values are used in all calculations of the partial results.

Description of the supporting files:

i) List of detected outlier values or outlier periods in the first homogenization cycle. – The filename has the form “PoutlieANAMEJJJ.res”.

“Poutlie” – constant
“ANAME” – dataset name
“JJJJ” – serial number of the candidate series
“.res” – constant

In the first homogenization cycle of ACMANT, the possible occurrences of monthly outlier values or outlier periods for 1-28 month sections of time series are examined, anything is the time resolution of the input data. However, the treatment of precipitation data differs from the base rule. In daily precipitation homogenization outlier filtering is not applied. In monthly precipitation homogenization only 1-month outlier values are examined, and only when user opts for the inclusion of outlier filtering. In the monthly homogenization of other climatic elements than precipitation, user may opt for no outlier filtering. In this case possible occurrences of 1-4 month outlier periods are not examined, but those of 5-28 month periods are still examined and treated.

For a network of *N* series, this file contains *N* blocks. For each block:

- line 1: serial number of time series within the network, number of detected outliers (*k’*), station identifier
- lines after 1 until 1+*k’*: serial number of outlier period, calendar year and month for the first month of the outlier period, length of outlier period in months.
- line *k’*+2: blank

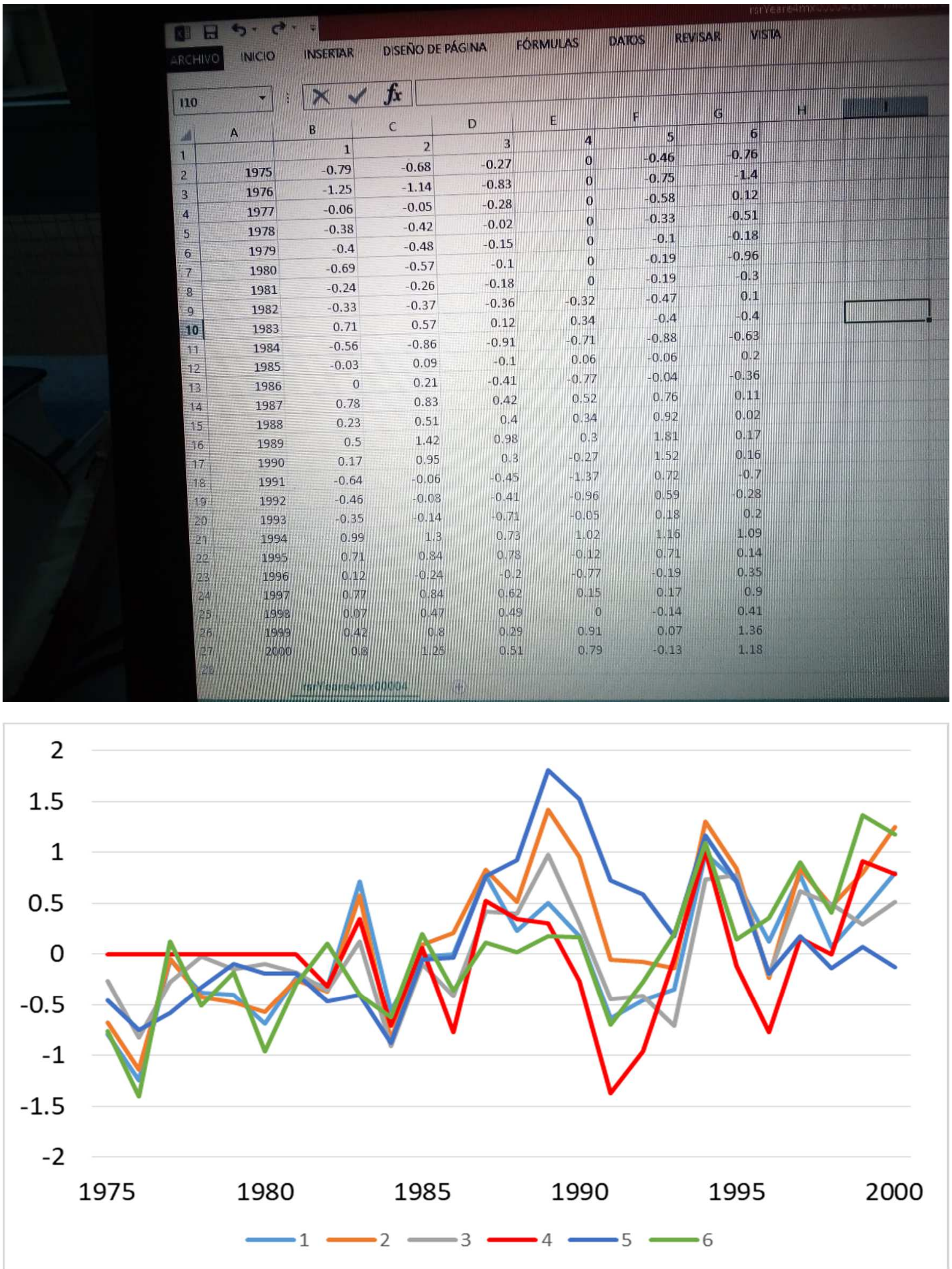
ii) Table of pairwise comparison scores. – The filename has the form “PpairwsANAMEJJJ.res”.

“Ppairws” – constant
“ANAME” – dataset name
“JJJJ” – serial number of the candidate series
“.res” – constant

This is the summary table of pairwise detection results without any combination of the pieces of the results and without considering metadata. The rows are for the years of the homogenized period of the candidate series, while the columns are for the time series of the examined network. The table shows that how many breaks were detected for each time series and each year. In ACMANT such sums can be fractions for the shared contribution of similarly effective function fittings of the break detection step. Symbol “…” is applied out of the homogenized period of time series.

iii) Anomalies of annual values relative to the long-term mean values, calculated from outlier filtered but non-homogenized values. – The filename has the form “rsrtypeANAMEJJJ.csv”

“rsr” – constant
“type” – “Year” for annual mean, “S-Wd” for summer - winter difference, “Rain” (“Snow”) for precipitation total of rainy (snowy) season
“ANAME” – dataset name
“JJJJ” – serial number of the candidate series
“.csv” – constant



	1	2	3	4	5	6
1975	-0.79	-0.68	-0.27	0	-0.46	-0.76
1976	-1.25	-1.14	-0.83	0	-0.75	-1.4
1977	-0.06	-0.05	-0.28	0	-0.58	0.12
1978	-0.38	-0.42	-0.02	0	-0.33	-0.51
1979	-0.4	-0.48	-0.15	0	-0.1	-0.18
1980	-0.69	-0.57	-0.1	0	-0.19	-0.96
1981	-0.24	-0.26	-0.18	0	-0.19	-0.3
1982	-0.33	-0.37	-0.36	-0.32	-0.47	0.1
1983	0.71	0.57	0.12	0.34	-0.4	-0.4
1984	-0.56	-0.86	-0.91	-0.71	-0.88	-0.63
1985	-0.03	0.09	-0.1	0.06	-0.06	0.2
1986	0	0.21	-0.41	-0.77	-0.04	-0.36
1987	0.78	0.83	0.42	0.52	0.76	0.11
1988	0.23	0.51	0.4	0.34	0.92	0.02
1989	0.5	1.42	0.98	0.3	1.81	0.17
1990	0.17	0.95	0.3	-0.27	1.52	0.16
1991	-0.64	-0.06	-0.45	-1.37	0.72	-0.7
1992	-0.46	-0.08	-0.41	-0.96	0.59	-0.28
1993	-0.35	-0.14	-0.71	-0.05	0.18	0.2
1994	0.99	1.3	0.73	1.02	1.16	1.09
1995	0.71	0.84	0.78	-0.12	0.71	0.14
1996	0.12	-0.24	-0.2	-0.77	-0.19	0.35
1997	0.77	0.84	0.62	0.15	0.17	0.9
1998	0.07	0.47	0.49	0	-0.14	0.41
1999	0.42	0.8	0.29	0.91	0.07	1.36
2000	0.8	1.25	0.51	0.79	-0.13	1.18

Figure 1. Upper panel: A *.csv data table for the 1975-2000 period of 6 time series.
Bottom panel: the figure of the data table by selecting Insert/Graphics/Lines

This table can be opened in Excel, and its purpose is to give users an easy access to make graphical comparisons between time series. In the data table the rows are for the years of the homogenized period of the candidate series, while the columns are for the time series of the examined network. When the homogenized period of a time series is shorter than the shown period, the non-interpreted data of that series are presented with zeros (e.g., series 4, 1975-1981 in Fig. 1). Notwithstanding, internal data gaps within homogenized periods are infilled by spatial interpolation.

This data table can be converted into figures within seconds, it is easy even for whom Excel is not a routine access. Figure 1 shows an example. For simplicity, a small network (6 time series) and a short period (1975-2000) are shown. For transforming a data table to figure, first select the data area (A – G; 1 – 27 in the upper panel of Fig. 1) with mouse, then choose Insert / Graphics / Lines from the menu. The result is shown in the bottom panel of Fig. 1. For larger networks and longer periods you may need the selection of one or more small sets of time series / subperiods. These all are very easy in Excel with moving rows and columns to the required order.

4.8. User editions in the small network (optional)

“Small network” (Sect. 4.4) is used in the homogenization of probability distribution (PDF). Any ACMANT generated small networks contain up to 11 time series (i.e., a candidate series and up to 10 neighbour series). When ACMANT finds 10 or more neighbour series with sufficient spatial correlation with the candidate series, the 10 best correlating series will be included in the small network, with an exception: when the homogenized period of a neighbour series is shorter than 20 years, and shorter also than the half of the length of the homogenized period of the candidate series, that neighbour series is excluded.

Users may change the set of the neighbour series. The number of neighbour series (S) must fulfil this condition: $3 \leq S \leq 15$. For the editions, ACMANT puts an editable file and a supporting file into directory Partial results.

The editable file is “SrlistJJJJ.res” where

“Srlist” – constant
“JJJJ” – serial number of the candidate series
“.res” – constant

Such files are only for one time series, also in 1-network homogenization, since the PDF homogenization is applied one-by-one for individual time series. In the edition of the network, the serial number of the selected neighbour series must be specified, while the other pieces of information are only for supporting the user. In multi-network homogenization the serial numbers are different from those in the original input dataset (Sect. 4.4). The first line of the file shows the key information if the serial numbers of the original dataset or those of the relevant subnetwork should be specified. The structure of the file:

- line 1: Key information about serial numbers
- line 2: Number of neighbour series (k)
- lines 3 to $2+k$, from left to right: serial number of time series within the small network, serial number of time series within the base network (or, in 1-network homogenization, within the original input dataset), spatial correlation with the candidate series and station identifier. An example:

```
Subnetwork used
 10
    1    2  0.819   Station 0004
    2    3  0.805   Station 0029
    3    6  0.614   Station 0026
    4    4  0.614   Station 0024
    5    7  0.591   Station 0012
    6    8  0.586   Station 0009
    7    5  0.584   Station 0015
    8    9  0.519   Station 0016
    9   10  0.516   Station 0019
   10   11  0.495   Station 0018
```

The serial numbers within base network are often close to an increasing series from 2 to 11 (1 = candidate series), since in constructing both the base network and small networks the rank order of the spatial correlations are used. However, some factors may modify this order, (length of neighbour series and the spatial correlations are recalculated in the ACMANT procedure with QC-d and homogenized data).

In the edited file the key information should be kept, the number of time series must be adjusted according to the number of selected neighbour series, and in the other lines only the serial number within the small network and that within the base network (or original input dataset) should be specified.

The supporting file for the small network edition is “Rlist.res”, this filename is constant. That file contains the base network neighbour series data in a similar format to that of the SrlistJJJJ.res. The Rlist.res file shows which neighbour series can be taken into consideration, their serial number within the base network (or within the original input dataset), their spatial correlations, their station identifier, and one more thing for each shown neighbour series: their homogenized period.

4.9. User editions for determining the reference HSP (optional)

In homogenization generally the default reference section of time series is the last homogeneous sub-period (HSP) of the time series. While this consensus practically never cause problems in the homogenization of the section means, it might damage observed data when a period of erroneous observations are applied as reference HSP in the homogenization of PDF. Therefore, the data quality of the last HSP should be controlled before applying an automatic homogenization of PDF. Further, the length of the reference HSP might also influence the quality of the PDF homogenization results, so that when inhomogeneities occur close to the end of the observation period, the use of an earlier and longer HSP as reference period could be advantageous.

In ACMANT, the PDF-homogenization is performed without seasonal division for variables FF, HH and PP, but season-specific homogenization is performed for TT and SS. Therefore, the editable file has different form according to the homogenized climatic variable.

i) PDF homogenization without seasonal division:

The editable file is HPDYEARJJJJ.res where

"HPDYEAR" – constant
"JJJJ" – serial number of the candidate series
".res" – constant

Line two shows the serial number of the HSP. Default value is $k + 1$ where k is the number of detected breaks with significant change in the probability distribution. (k breaks divides the homogenized period to $k + 1$ homogeneous sub-periods, and the default serial number is always the last HSP.) Below the serial number of reference HSP the file shows the homogenized period and its division to HSPs showing the date of the last day for each HSP. Users may change the serial number of reference HSP to a value between 1 and k +1, and they should not change the other parts of the file.

ii) PDF homogenization with separation according to seasons within the procedure

The editable file is HPDMONTHSJJJJ.res where

"HPDMONTHS" – constant
"JJJJ" – serial number of the candidate series
".res" – constant

In this file month-specific serial numbers of reference HSPs are shown. The first line of the file shows the homogenized period, and the serial numbers for individual calendar months are presented in lines 3 to 14. Below them, the divisions to HSPs are shown again for individual months. An example excerpt of file is shown here:

Homogenized period: 1951 - 2005
Serial numbers of ref. HSPs:
Jan   1
Feb   2
Mar   4
Apr   4
May   4
Jun   3
Jul   3

Aug 2
Sep 2
Oct 2
Nov 1
Dec 1

Jan
Jan 2005 12 31

Feb
Feb 1979 1 31
Feb 2005 12 31

Mar
Mar 1966 3 31
Mar 1975 11 30
Mar 1979 1 31
Mar 2005 12 31

When the default serial number is 1, it means that no significant break was found in the probability distribution. In the shown example, there is nothing to do with the data of January, but, for instance, users may choose between four HSPs in homogenizing the PDF for March. Users should not make editions in other parts of the file than the serial numbers for reference HSPs in lines 3 to 14.

Note when no significant break occurs throughout the year, this file is not generated.

4.10. Possibility to repeat the homogenization of a time series

When the interactive homogenization of a time series has been finished in multi-network homogenization, ACMANT asks if the user accepts the homogenization results. For the decision of accepting or not accepting the homogenization results, user can examine all the partial results and the requested other output items, being every result in the directory Output at this phase. If user accepts the results, he/she types “A” (accept), and the homogenization continues with the subsequent network. If user is unsatisfied with the results, he/she may edit the network content, or may prepare a homogenization with differing editions, and when he/she is ready with these, types “R” (repeat). Then ACMANT will repeat the homogenization of the network of that candidate series. This repetition option makes easy to treat possible visible problems of the homogenization results immediately, even for large datasets.

5. Results of the ACMANTv6 homogenization procedure

The output may include various versions of homogenized series, a summary of the detected breaks and outliers, files with monthly and daily reliability indicators related to the data treatment during the homogenization procedure, the list of neighbour series and their spatial correlations with the candidate series, one or more control messages and warning messages, and some partial results. All these depend on the user's choices provided at the beginning of the homogenization procedure (Sect. 4.2). One easy option is to choose the default output package, its content is detailed in Sect. 5.1 and 5.2. However, users may choose other output items than which belong to the default output package. The optional output items are described in Sect. 5.3. Note that the partial results were described in Sect. 4.5.

5.1. Default output package of monthly homogenization

i) Homogenized monthly series with gap filling within the homogenized period (for possibly existing missing data of the raw series). Their names have the form of "ANAMEJJJJt.txt", where

"ANAME" – name of the dataset
"JJJJ" – serial number of the time series
"t.txt' – constant

This file contains the homogenized time series, but note that the requested homogenized data tables are always generated, even when no section of a time series has been homogenized. So that the number of files of this kind is the same as the number of time series in the input dataset. The data format is exactly the same as the input data format.

Note that monthly data tables can be requested also when the input dataset is of daily resolution. In this case, output monthly data tables present monthly mean values for the majority of climate variables, but monthly totals for precipitation (RR), sunshine duration and radiation (SS).

ii) Summary of the homogenization procedure with the list of the detected breaks and outliers and some other characteristics. The name of this file has the form of "ANAME_breaks.txt".

"ANAME" – name of the dataset
"_breaks.txt" – constant

The organisation of the data in this file is as follows: The list comprises *N* sub-lists, where *N* stands for the number of stations. A sub-list generally contains the following 3 parts: a) headline, b) list of breaks, c) list of outliers. The headline contains (from the left to the right) the serial number of the time series, the starting and ending years of the treated period (Sect. 3.5), the starting and ending years of the homogenized period, the

number of detected breaks, the number of detected outliers (only in outlier filtering "yes" mode), and the station identifier. In part (b), the properties of the detected breaks are presented. They are the serial number of break, the year and month of break position and the break size for the annual mean of the tested variable. In bivariate homogenization (Sect. 4.5) two break sizes are shown, one for each variable. In part (c), the properties of the detected outlier periods are presented, they are the serial number of the outlier period, the year and month of the last month of the outlier period, the duration of outlier period (in months) and the magnitude of deviation. Some notes:
- If no break is detected in the time series, there is no b) part in the sub-list.
- If no outlier is detected in the time series, or outlier filtering "No" mode is applied, there is no c) part in the sub-list.
- The most frequent length of outlier periods is 1 month (i.e. one single outlier value), but lengths may vary between 1 and 4 months. In the partial results the maximum length of outlier periods is 28 months (Sect. 4.5). However, in the final homogenization cycle outlier periods of longer than 4 months are treated as a pair of breaks, and their properties are shown in the break list (part b).
- If homogenization has not been performed, character "0" is written into the places of the starting and ending years of the homogenized period, and "–1" into the place of the number of breaks.
- If a time series does not have treated period, "0" are shown in the places of the starting and ending years of the treated period.

iii) List of neighbour series for each candidate series, and the spatial correlations between candidate series and their neighbour series. The name of the file has the form of "ANAME_rlist.txt".

"ANAME" – name of the dataset
"_rlist.txt" – constant

In multi-network homogenization this file shows the network structures and the correlations between the candidate series and neighbour series. The network structures are shown both for the base network and the extended network. In 1-network homogenization (for small datasets), the candidate series – neighbour series relations are also shown for every time series, since the roles of being candidate series or neighbour series change during the homogenization procedure. Note that when a candidate series does not have homogenized section, no neighbour series or spatial correlation is shown for that. Furthermore, when a neighbour series cannot be homogenized, that series is not shown in any neighbour series list, even if it was selected for some candidate series during the automatic network construction. The form of data is as follows: for each candidate series with non-zero homogenized period has a headline, followed by the list of neighbour series ordered according to decreasing spatial correlations. The headline includes (from the left to the right) the serial number of the candidate series, the starting and ending years of its homogenized period, and its station identifier. Then, in each line after the headline, the serial number of correlation value, the correlation value itself and the station identifier of the neighbour series are shown.

In the process of automatic networking the first *K* series of the *N* series of the extended network compose the base network (Sect. 4.4). Although the spatial correlation is the most important factor of the selection of neighbour series in the automatic networking, the rank order of the neighbour series may change during the homogenization. For indicating the series which belong only to the extended network, letter “A” is added to their serial numbers.

iv) Control message. In running ACMANT, “Control message.txt” file is generated. If no problem has been detected during the run of the program, “….homogenization has been completed” will be the message. However, other messages are possible. In multi-network homogenization, at least one message is provided for each network homogenization.

v) Warning message. When the program finds synchronous breaks for high ratio of time series, it generates warning messages. The ratio of synchronous breaks is controlled for each candidate series and each year of the period. The threshold ratio for generating warning messages depends both on the number of compared series and the phase of the homogenization. The program controls the synchronous breaks in three steps of the homogenization procedure: a) in the metadata check, at the beginning, b) in the statistically detected breaks of the first phase of homogenization, c) in the overall number of breaks (statistically detected + metadata), in the final homogenization cycle. Often, several warning messages are generated for the same year, as if synchronous breaks occur, it may be present for several candidate series and in different phases of the homogenization.

The form of the output filename is constant, “Warning_coinc_breaks.txt”, and it appears in the output list when at least 1 warning message has been generated. Each line of the file is a separate warning message. From left to right, the phase of the homogenization, the serial number of candidate series, calendar year, number of compared time series and the number of time series with synchronous breaks stand. In 1-network homogenization, any warning message is common for all the time series, and the string “_ntw” stands in the place of the serial number of candidate series. In bivariate homogenization, separate warning messages may be generated for the two variables in the phase of pre-homogenization, and thus the warning message may end with “Year” (annual mean) “S-Wd” (summer-winter difference), “Rain” (mean for the rainy season) “Snow” (mean for the snowy season).

5.2. Default output package of daily homogenization

v) Homogenized daily series with gap filling within the homogenized period. Their names have the form of “ANAMEJJJJv.txt” where

“ANAME” – name of the dataset
“JJJJ” – serial number of the time series
“v.txt” – constant

Its data format is exactly the same as the input daily data format.

The items ii) - iv) of the standard monthly homogenization output (Sect. 5.1) are included also in the standard daily homogenization outputs, however the content of "ANAME_breaks.txt" (item ii) slightly differs:
- In daily homogenization, break positions are shown with daily preciseness. Note that for low signal-to-noise ratio on the daily scale (which is frequent), the calendar day for the break position is the default value, i.e. the last day of the month.
- For outlier periods, the starting and ending dates are presented with daily preciseness, instead of their length in months.
- In daily precipitation homogenization no outlier filtering is performed, and thus such kind of results are not presented
- Sometimes missing data code (-999.9) stands in the place of the deviation (systematic bias) size of the outlier period. It is either because the detection results do not prove a platform shaped structure (which is necessary to estimate the mean systematic bias), or for the lack of sufficient observed data in neighbour series to provide reliable estimations.

5.3. Optional output items: time series

Several optional output items are data tables of homogenized time series or tables of reliability indicators. All for these files have the same file format as that of "ANAMEJJJJt.txt" (for monthly data) or "ANAMEJJJJv.txt" (for daily data). The letter before ".txt" shows the kind of the output item, the other characters of the filenames are the same for a given time series.

vi) ANAMEJJJJs.txt – homogenized monthly time series without gap filling for missing data.

vii) ANAMEJJJJh.txt – homogenized monthly time series completed with interpolated data to the target period when data gaps occur in the input data.

viii) ANAMEJJJJu.txt – homogenized daily time series without gap filling for missing data.

ix) ANAMEJJJJg.txt – homogenized daily time series completed with interpolated data to the target period when data gaps occur in the input data.

x) ANAMEJJJJj.txt – Monthly data table of reliability indicators. The data format is the same as for climate data tables, except that whole numbers (0…9) of reliability indications stand on the places of climate data of climate data tables. The meanings of reliability indicators are presented in Sect. 5.4.

xi) ANAMEJJJJi.txt – Daily data table of reliability indicators. The data format is the same as for climate data tables, except that numbers of reliability indications stand on the places of climate data of climate data tables.

5.4. Meaning of reliability indicators

Although the file ANAME_breaks.txt shows the homogenized period for each time series, reliability indicators might show more about the reliability of the homogenized data for two reasons:

- When less than three time series have data for a specific year, that year is excluded from the homogenization for all the stations. Such years might occur either within or out of the homogenized period.

- ACMANT automatically performs gap filling with spatial interpolation, but when the number of neighbour series or the spatial correlations are low, the confidence of interpolated values is limited.

Generally the lower values mean higher reliability, except for the 0 code.

0 – sporadic observed value, out of the treated period of the time series, or observed value in a section of the candidate series where the number of neighbour series with observed values is 0 or 1.
1 – observed value within the homogenized period
2 – observed value out of the homogenized period
3 – Interpolated value from at least 4 sufficiently correlating neighbour series.
4 – Interpolated value from 3 highly correlating neighbour series, or from more than 3 but only moderately correlating neighbour series.
5 – Interpolated value from at least 3 fairly correlating neighbour series.
6 – Interpolated value from 2 fairly correlating neighbour series, or from more but poorly correlating neighbour series.
7 – Interpolated value by the use of 1 only neighbour series.
8 – Missing value is substituted with the climatic normal value due to the complete lack of observed values in neighbour series. Frequent occurrence of this code number may be present when the target period is longer than the periods of observed data, and full completion of time series is performed. When no data is available for spatial interpolation, the climatic normal values are included (repeatedly) in the artificially lengthened series. Note, however, that this code sometimes occur within the homogenized period due to synchronous data gaps (e.g. for political events).
9 – missing data for which interpolation has not been performed. This code may occur out of the homogenized period, when data tables of time series completed to the target period are not requested.

Note: The coding between 3 and 7 is based on the number of neighbour series, the spatial correlations between series, the frequency of comparable data pairs around the date of the interpolation, and if the data are within the homogenized period yes or not. Due to this complexity, not all details of the coding rules are presented.

5.5. Lists and tables of calculation results

These tables are presented (a) for time series which is subjected to interactive homogenization always, (b) for all time series when user requires these results, and answers positively at Sect. 4.2 (xv).

Most of the calculation results are the same as which are shown in directory Partial results during interactive homogenization, although not the whole content of Partial results is saved, and the format of the final results sometimes differ from that of the partial results. One item is fully new here, and it is the tables of annual anomaly series for homogenized time series.

(xii) Anomalies of annual values relative to the long-term mean values, calculated from outlier filtered but non-homogenized values.

In bivariate homogenization (Sect. 4.7) it is presented to both variables, otherwise only to the annual means. The filename and file content is exactly the same as the partial results of item (iii) of Sect. 4.7.

(xiii) Anomalies of annual values relative to the long-term mean values, calculated from the final homogenized values. The filename has the form of "hsrtypeANAMEJJJ.csv" where

"hsr" – constant
"type" – "Year" for annual mean, "S-Wd" for summer - winter difference, "Rain" ("Snow") for precipitation total of rainy (snowy) season
"ANAME" – dataset name
"JJJJ" – serial number of the candidate series
".csv" – constant

The format of the file is the same as for type (xii).

(xiv) List of detected breaks in the first homogenization cycle

This item has two alternative forms according to interactive homogenization of the candidate series was performed yes or no. When the homogenization was automatic, the filename, file content and file format are the same as those of the editable file "PBRtypeANAMEJJJJ.res" of Sect. 4.7. When interactive homogenization was performed, the list of detected breaks is written twice for each series (either user editions were performed or not): first the automatically generated list, and thereafter the user edited list are shown. An example showing the results for the first 3 time series of a network:

```
  1  1951 2005   0003
   2
  1  1975    0.24
  2  1978    0.43
   1
  1  1975
```

```
  2  1955 2004   0009
   3
  1  1963   -3.00
  2  1965   -1.92
  4  1979   -1.18
   3
  1  1963
  2  1965
  4  1979

3  1961 2004   0004
   0
   1
  1  1989
```

The example shows that user deleted a break of the first series, left the results of the second series unchanged, and introduced a break to the third series. The example illustrates that user never defines break size.

xv) List of detected outliers in the first homogenization cycle. It is the same as support file (i) of Sect. 4.7.

xvi) Table of pairwise comparison scores. It is the same as support file (ii) of Sect. 4.7.

xvii) Summary table of small networks. The filename is ANAME_Srlist.txt where

"ANAME" – dataset name
"_Srlist.txt" – constant

This file contains all small networks generated during a homogenization procedure. For each of such networks, the list of neighbour series and their spatial correlation with the candidate series are shown in a similar format to files "SrlistJJJJ.res". The only difference from those lists is that serial numbers within the base network are not shown in the final results, since the time series are identifiable by the shown station identifier. User may edit "SrlistJJJJ.res", and when they apply changes, the user edited versions are shown in the output.

xviii) Summary table of reference HSPs in the PDF homogenization for variables FF, HH or PP. The filename has the form of "ANAMEHpdYEAR.txt" where

"ANAME" – dataset name
"HpdYEAR.txt" – constant

This file shows results for each candidate series with at least 1 significant break of the probability distribution. The results are shown in two lines for each series: Line 1 shows the serial number of time series, its homogenized period and the station identifier. Line 2 shows the serial number of the reference HSP and its duration. When user changes the reference HSP in interactive homogenization, the user edited version is shown.

xix) Summary table of reference HSPs in the PDF homogenization for variables TT or SS. The filename has the form of “ANAMEHpdMONTHS.txt” where

“ANAME” – dataset name
“HpdMONTHS.txt” – constant

The content and format are partly similar to that of (xviii), but the serial number and duration of a reference HSP are shown separately to each calendar month. Therefore the results for a given candidate series are shown in 1+12 lines instead of 2 lines.

5.6. Documentation of user interventions

In the present ACMANT version not every user intervention is documented. In fact both of the original and edited versions are shown only for the break detection results of the first homogenization cycle (item xiv). So that if user would like to compare the original and edited files after homogenization, he/she should save the original version manually.

Another issue is the documentation of repeated homogenization for a given candidate series. Some result files collect the pieces of the results for each candidate series and each homogenization exercise (if repetitions occur). There the results of the individual homogenization exercises are documented. The files with summary tables are:
- item (ii) Final lists of treated periods, homogenized periods, detected breaks and outliers;
- item (iii) List of the neighbour series and their spatial correlation with the candidate series:
- item (iv) Control messages;
- item (v) Warning messages;
- item (xvii) Summary table of small networks;
- items (xviii-xix) Summary table of reference HSPs.

6. Possible technical problems with using ACMANT

Although all programs of the software have been tested, errors still might occur.

The following symptoms indicate that the program failed during the homogenization of a candidate series: (i) In the list of control messages (“Control message.txt”) a candidate series is missing; (ii) Some of the basic output items or all the output items of a given candidate series are missing; (iii) When the missing of an output item is unexpected for logical connections; (iv) Directory Output or directory Works include error message from the program (the homogenization may stop with or without giving error message). Note that the homogenization of some candidate series might be denied for the insufficient spatial-temporal coherence of the data, but this is not a software error, and in this case control message and some other output items are still generated.

When a homogenization procedure failure is realised, first the input data should be checked, as errors in the input data preparation may cause stop of the homogenization. In such cases, likely an error message from the program will be present in directory Works. A relatively frequent data preparation error is that the same time series occurs repeatedly, under different station names. In this case, the homogenization will stop with the error message of “Indefinite equation system”.

If ACMANT fails to homogenize your data, and irregularities of the input dataset cannot be found, then please report the error to me.